

G6 Constructing differentiation equations

Name: _____

Class: _____

Date: _____

Time: **310 minutes**

Marks: **259 marks**

Comments:

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Q1.

A market trader notices that daily sales are dependent on two variables:

number of hours, t , after the stall opens

total sales, x , in pounds since the stall opened.

The trader models the rate of sales as directly proportional to $\frac{8-t}{x}$

After two hours the rate of sales is £72 per hour and total sales are £336

(a) Show that

$$x \frac{dx}{dt} = 4032(8-t)$$

(3)

(b) Hence, show that

$$x^2 = 4032t(16-t)$$

(3)

(c) The stall opens at 09.30.

(i) The trader closes the stall when the rate of sales falls below £24 per hour.

Using the results in parts (a) and (b), calculate the earliest time that the trader closes the stall.

(6)

(ii) Explain why the model used by the trader is not valid at 09.30.

(2)

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(Total 14 marks)

Q2.

Sam goes on a diet. He assumes that his mass, m kg after t days, decreases at a rate that is inversely proportional to the cube root of his mass.

(a) Construct a differential equation involving m , t and a positive constant k to model this situation.

(3)

(b) Explain why Sam's assumption may not be appropriate.

(1)

(Total 4 marks)

Q3.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A .

- (a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

- (b) When it was first noticed, the weed covered an area of 0.25 m^2 . Twenty days later the weed covered an area of 0.5 m^2

- (i) State the value of B .

(1)

- (ii) Show that the model for the area covered by the weed can be written as

$$A = 2^{\frac{t}{20} - 2}$$

(4)

- (iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

- (c) State one limitation of the model.

(1)

- (d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

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Q4.

The height of the tide in a certain harbour is h metres at time t hours. Successive high tides occur every 12 hours.

The **rate of change** of the height of the tide can be modelled by a function of the form $a \cos(kt)$, where a and k are constants. The largest value of this rate of change is 1.3 metres per hour.

Write down a differential equation in the variables h and t . State the values of the constants a and k .

(Total 3 marks)

Q5.

- (a) A water tank has a height of 2 metres. The depth of the water in the tank is h metres

at time t minutes after water begins to enter the tank. The rate at which the depth of the water in the tank increases is proportional to the difference between the height of the tank and the depth of the water.

Write down a differential equation in the variables h and t and a positive constant k .

(You are not required to solve your differential equation.)

(3)

- (b) (i) Another water tank is filling in such a way that t minutes after the water is turned on, the depth of the water, x metres, increases according to the differential equation

$$\frac{dx}{dt} = \frac{1}{15x\sqrt{2x-1}}$$

The depth of the water is 1 metre when the water is first turned on.

Solve this differential equation to find t as a function of x .

(8)

- (ii) Calculate the time taken for the depth of the water in the tank to reach 2 metres, giving your answer to the nearest 0.1 of a minute.

(1)

(Total 12 marks)

Q6.

A giant snowball is melting. The snowball can be modelled as a sphere whose surface area is decreasing at a constant rate with respect to time. The surface area of the sphere is A cm² at time t days after it begins to melt.

- (a) Write down a differential equation in terms of the variables A and t and a constant k , where $k > 0$, to model the melting snowball.

(2)

- (b) (i) Initially, the radius of the snowball is 60 cm, and 9 days later, the radius has halved.

Show that $A = 1200\pi(12 - t)$.

(You may assume that the surface area of a sphere is given by $A = 4\pi r^2$, where r is the radius.)

(4)

- (ii) Use this model to find the number of days that it takes the snowball to melt completely.

(1)

(Total 7 marks)

Q7.

A botanist is investigating the rate of growth of a certain species of toadstool. She observes that a particular toadstool of this type has a height of 57 millimetres at a time 12 hours after it begins to grow.

$$h = A \left(1 - e^{-\frac{1}{4}t} \right)$$

She proposes the model, where A is a constant, for the height h millimetres of the toadstool, t hours after it begins to grow.

(a) Use this model to:

(i) find the height of the toadstool when $t = 0$;

(1)

(ii) show that $A = 60$, correct to two significant figures.

(2)

$$h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$$

(b) Use the model to:

(i) show that the time T hours for the toadstool to grow to a height of 48 millimetres is given by

$$T = a \ln b$$

where a and b are integers;

(3)

(ii) show that $\frac{dh}{dt} = 15 - \frac{h}{4}$;

(3)

(iii) find the height of the toadstool when it is growing at a rate of 13 millimetres per hour.

(1)

(Total 10 marks)

Q8.

(a) Solve the differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

given that $x = 80$ when $t = 0$. Give your answer in the form $x = f(t)$.

(6)

(b) A fungus is spreading on the surface of a wall. The proportion of the wall that is unaffected after time t hours is $x\%$. The rate of change of x is modelled by the

differential equation

$$\frac{dx}{dt} = -\frac{1}{5}(x+1)^{\frac{1}{2}}$$

At $t = 0$, the proportion of the wall that is unaffected is 80%. Find the proportion of the wall that will still be unaffected after 60 hours.

(2)

- (c) A biologist proposes an alternative model for the rate at which the fungus is spreading on the wall. The total surface area of the wall is 9 m^2 . The surface area that is **affected** at time t hours is $A \text{ m}^2$. The biologist proposes that the rate of change of A is proportional to the product of the surface area that is affected and the surface area that is unaffected.

- (i) Write down a differential equation for this model.

(You are not required to solve your differential equation.)

(2)

- (ii) A solution of the differential equation for this model is given by

$$A = \frac{9}{1 + 4e^{-0.09t}}$$

Find the time taken for 50% of the area of the wall to be affected. Give your answer in hours to three significant figures.

(4)

(Total 14 marks)

Q9.

- (a) The number of fish in a lake is decreasing. After t years, there are x fish in the lake. The rate of decrease of the number of fish is proportional to the number of fish currently in the lake.

- (i) Formulate a differential equation, in the variables x and t and a constant of proportionality k , where $k > 0$, to model the rate at which the number of fish in the lake is decreasing.

(2)

- (ii) At a certain time, there were 20 000 fish in the lake and the rate of decrease was 500 fish per year. Find the value of k .

(2)

- (b) The equation

$$P = 2000 - Ae^{-0.05t}$$

is proposed as a model for the number of fish, P , in another lake, where t is the time in years and A is a positive constant.

On 1 January 2008, a biologist estimated that there were 700 fish in this lake.

- (i) Taking 1 January 2008 as $t = 0$, find the value of A .

(1)

- (ii) Hence find the year during which, according to this model, the number of fish in this lake will first exceed 1900.

(4)

(Total 9 marks)

Q10.

A disease is spreading through a colony of rabbits. There are 5000 rabbits in the colony. At time t hours, x is the number of rabbits infected. The rate of increase of the number of rabbits infected is proportional to the product of the number of rabbits infected and the number not yet infected.

- (a) (i) Formulate a differential equation for $\frac{dx}{dt}$ in terms of the variables x and t and a constant of proportionality k .

(2)

- (ii) Initially, 1000 rabbits are infected and the disease is spreading at a rate of 200 rabbits per hour. Find the value of the constant k .

(You are **not** required to solve your differential equation.)

(2)

- (b) The solution of the differential equation in this model is

$$t = 4 \ln \left(\frac{4x}{5000 - x} \right)$$

- (i) Find the time after which 2500 rabbits will be infected, giving your answer in hours to one decimal place.

(2)

- (ii) Find, according to this model, the number of rabbits infected after 30 hours.

(4)

(Total 10 marks)

Q11.

A particle, of mass 12 kg, is moving along a straight horizontal line. At time t seconds, the particle has speed v m s⁻¹. As the particle moves, it experiences a resistance force of magnitude $4v^{\frac{1}{3}}$. No other horizontal force acts on the particle.

The initial speed of the particle is 8 m s⁻¹.

- (a) Show that

$$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$$

(6)

- (b) Find the value of t when the particle comes to rest.

(1)

(Total 7 marks)

Q12.

A car accelerates from rest along a straight horizontal road.

The car's engine produces a constant horizontal force of magnitude 4000 N.

At time t seconds, the speed of the car is $v \text{ m s}^{-1}$, and a resistance force of magnitude $40v$ newtons acts upon the car.

The mass of the car is 1600 kg.

- (a) Show that $\frac{dv}{dt} = \frac{100 - v}{40}$.

(2)

- (b) Find the velocity of the car at time t .

(6)

(Total 8 marks)

Q13.

Alice places a toy, of mass 0.4 kg, on a slope. The toy is set in motion with an initial velocity of 1 m s^{-1} down the slope. The resultant force acting on the toy is $(2 - 4v)$ newtons, where $v \text{ m s}^{-1}$ is the toy's velocity at time t seconds after it is set in motion.

- (a) Show that $\frac{dv}{dt} = -10(v - 0.5)$.

(2)

- (b) By using $\int \frac{1}{v-0.5} dv = - \int 10 dt$, find v in terms of t .

(5)

- (c) Find the time taken for the toy's velocity to reduce to 0.55 m s^{-1} .

(3)

(Total 10 marks)

Q14.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s^{-1} .



At time t seconds after it is projected, the stone has speed v m s⁻¹ and it experiences a resistance force of magnitude $9.8v$ newtons.

- (a) When $t \geq 0$, show that

$$\frac{dv}{dt} = -1.96(v - 5)$$

(2)

- (b) Find v in terms of t .

(5)

(Total 7 marks)

Q15.

Vicky has mass 65 kg and is skydiving. She steps out of a helicopter and falls vertically. She then waits a short period of time before opening her parachute. The parachute opens at time $t = 0$ when her speed is 19.6 m s⁻¹, and she then experiences an air resistance force of magnitude $260v$ newtons, where v m s⁻¹ is her speed at time t seconds.

- (a) When $t > 0$:

- (i) show that the resultant downward force acting on Vicky is

$$65(9.8 - 4v) \text{ newtons}$$

(1)

- (ii) show that $\frac{dv}{dt} = -4(v - 2.45)$.

(2)

- (b) By showing that $\int \frac{1}{v - 2.45} dv = -\int 4 dt$, find v in terms of t .

(5)

(Total 8 marks)

Q16.

A car, of mass m kg, is moving along a straight horizontal road. At time t seconds, the car has speed v m s⁻¹. As the car moves, it experiences a resistance force of magnitude $2mv^{\frac{5}{4}}$ newtons. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -2v^{\frac{5}{4}}$$

(1)

- (b) The initial speed of the car is 16 m s⁻¹.

Show that

$$v = \left(\frac{2}{t+1} \right)^4$$

(5)

(Total 6 marks)

Q17.

A golf ball, of mass m kg, is moving in a straight line across smooth horizontal ground. At time t seconds, the golf ball has speed v m s⁻¹. As the golf ball moves, it experiences

a resistance force of magnitude $0.2mv^{\frac{1}{2}}$ newtons until it comes to rest. No other horizontal force acts on the golf ball.

Model the golf ball as a particle.

(a) Show that

$$\frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

(1)

(b) When $t = 0$, the speed of the golf ball is 16 m s⁻¹.

Show that $v = (4 - 0.1t)^2$.

(5)

(c) Find the value of t when $v = 1$.

(3)

(d) Find the distance travelled by the golf ball as its speed decreases from 16 m s⁻¹ to 1 m s⁻¹.

(4)

(Total 13 marks)

Q18.

A particle is moving along a straight line. At time t , the velocity of the particle is v .

$$-\frac{\lambda}{v^{\frac{1}{4}}}$$

The acceleration of the particle throughout the motion is $-\frac{\lambda}{v^{\frac{1}{4}}}$, where λ is a positive constant. The velocity of the particle is u when $t = 0$.

Find v in terms of u , λ and t .

(Total 7 marks)

Q19.



A stone, of mass 0.05 kg, is moving along the smooth horizontal floor of a tank, which is filled with oil. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude $0.08 v^2$.

- (a) Show that

$$\frac{dv}{dt} = -1.6v^2$$

(2)

- (b) The initial speed of the stone is 3 m s^{-1} .

Show that

$$v = \frac{15}{5 + 24t}$$

(5)

(Total 7 marks)

Q20.

A stone, of mass m , is moving in a straight line along smooth horizontal ground.

At time t , the stone has speed v . As the stone moves, it experiences a total resistance force of magnitude $\lambda m v^{\frac{3}{2}}$, where λ is a constant. No other horizontal force acts on the stone.

- (a) Show that

$$\frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

(2)

- (b) The initial speed of the stone is 9 m s^{-1} .

Show that

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

(7)

- (c) Find, in terms of λ , the time taken for the speed of the stone to drop to 4 m s^{-1} .

(3)

(Total 12 marks)

Q21.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has

speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s^{-1} .

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s^{-1} .

(3)

- (d) Find, in terms of m , the work done by the force in slowing the car from 20 m s^{-1} to 10 m s^{-1} .

(3)

(Total 11 marks)

Q22.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s^{-1} .

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s^{-1} .

(3)

(Total 8 marks)

Q23.

A stone of mass m is moving along the smooth horizontal floor of a tank which is filled with a viscous liquid. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude λmv , where λ is a constant.

- (a) Show that

$$\frac{dv}{dt} = -\lambda v$$

(2)

- (b) The initial speed of the stone is U .

Show that

$$v = Ue^{-\lambda t}$$

(4)

(Total 6 marks)

Q24.

A car, of mass 1600 kg, is travelling along a straight horizontal road at a speed of 20 m s^{-1} when the driving force is removed. The car then freewheels and experiences a resistance force. The resistance force has magnitude $40v$ newtons, where $v \text{ m s}^{-1}$ is the speed of the car after it has been freewheeling for t seconds.

Find an expression for v in terms of t .

(Total 7 marks)

Q25.

A student is modelling the motion of a small boat as it moves on a lake. When the speed of the boat is 12 m s^{-1} , the engine is switched off. At time t seconds later, it has a velocity of $v \text{ m s}^{-1}$ and experiences a resistance force of magnitude $20v$ newtons. The mass of the boat is 80 kg.

To set up a simple model for the motion of the boat, the student assumes that the water in the lake is still and that the boat travels in a straight line.

- (a) Explain how these two assumptions allow the student to create a simple model.

(2)

- (b) State one other assumption that the student should make.

(1)

- (c) (i) Express $\frac{dv}{dt}$ in terms of v .

(2)

- (ii) Find an expression for v in terms of t .

(5)

(Total 10 marks)

Q26.

A particle of mass 20 kg moves along a straight horizontal line. At time t seconds the

velocity of the particle is $v \text{ m s}^{-1}$. A resistance force of magnitude $10\sqrt{v}$ newtons acts on the particle while it is moving. At time $t = 0$ the velocity of the particle is 25 m s^{-1} .

- (a) Show that, at time t

$$v = \left(\frac{20 - t}{4} \right)^2$$

(7)

- (b) State the value of t when the particle comes to rest.

(1)

(Total 8 marks)

Q27.

A particle travels along a straight line. At time t , the velocity of the particle is v . When $t = 0$,

$v = u$. The acceleration of the particle throughout the motion is $\frac{k}{v}$, where k is a constant.

Show that

$$v^2 = u^2 + 2kt$$

(Total 6 marks)

Q28.

John drops a rock of mass $m \text{ kg}$ from the top of a vertical cliff. At time t seconds after the rock is dropped, it has speed $v \text{ m s}^{-1}$ and the air resistance to its motion is of magnitude $0.1mv$ newtons.

- (a) Show that the differential equation connecting v and t for this motion is $\frac{dv}{dt} = g - 0.1v$.

(2)

- (b) Solve this differential equation to find v in terms of g and t .

(5)

- (c) Hence show that the speed of the rock cannot exceed 98 m s^{-1} .

(1)

(Total 8 marks)

Q29.

A car accelerates from rest along a straight horizontal road. It experiences a constant horizontal forward force of magnitude 2000 newtons and a resistance force. The resistance force has magnitude $40v$ newtons, when the speed of the car is $v \text{ m s}^{-1}$. The mass of the car is 1000 kg.

- (a) Show that

$$\frac{dv}{dt} = \frac{50 - v}{25}$$

(2)

- (b) Find the velocity of the car at time t .

(5)

(Total 7 marks)

Q30.

A particle of mass m is moving along a straight horizontal line. At time t the particle has speed v . Initially the particle is at the origin and has speed U . As it moves the particle is subject to a resistance force of magnitude mkv^3 .

- (a) Show that $v^2 = \frac{U^2}{2kU^2t + 1}$.

(6)

- (b) What happens to v as t increases?

(1)

(Total 7 marks)



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